

05 Demand Shape & Load Factor

System demand MW · load factor · demand response value · grid intelligence

WHAT TO FIND IMMEDIATELY

Load factor	Average demand / peak demand as %. High (>75%) = flat, efficient profile. Low (<60%) = peaky, stresses the grid, expensive to serve.
Evening peak time	When exactly does demand peak? Is it shifting later week-on-week? EV charging is a structural force pushing the peak later.
Midday dip	A visible net demand dip at 12:00–14:00 on sunny days = rooftop solar reducing grid demand. The duck curve forming in Ireland.
Demand response signal	Unusual sharp drops during high-price periods that don't match normal shape = industrial DR activating. Worth flagging when visible.

ANGLES — ROTATE ONE PER POST

Load factor headline	Today's load factor of [X]% is [the lowest/highest] this month. A [peaky/flat] day that [stresses the grid / is cheap to serve].
DR value calculation	If a 1MW industrial customer shifted [N] hours of load off the evening peak, they would have saved €[X] at today's DAM prices. Quantify it.
Duck curve emerging	Is there a visible midday demand dip? How deep is it? Track this monthly — as rooftop solar grows, the curve gets more pronounced.
Day type analysis	Weekday vs weekend demand shapes are distinct. Saturday is flat; Monday morning ramp is steep. Call out anomalies against the expected shape.
Demand vs price mismatch	High demand but low prices (lots of wind) — or low demand but high prices (no wind, thermal price-setting). These mismatches tell the supply story.
Network planning lens	The peak demand value and its timing determines grid investment decisions. A peak at 17:30 vs 20:30 requires different network solutions.

HOOK SENTENCE FORMULA

Load factor	"Today's load factor of [X]% is the [lowest/highest] this [week/month] — a [peaky/flat] profile that [stresses the grid / rewards flexible assets]."
DR value	"A 1MW industrial customer shifting just 2 hours of load off yesterday's evening peak would have saved approximately €[X] at DAM prices."
Duck curve	"The midday demand dip reached [X]MW at [time] — [deeper/shallower] than last week. Ireland's duck curve is [forming/deepening]."

WHY THIS MATTERS FOR SALARY

Demand analysis is the core skill for grid planning, network investment, demand response, and smart metering roles. Publishing load factor and DR value analysis signals that you think about the grid as a system — not just a price signal. This is the angle that separates market analysts from energy systems analysts, who typically command higher salaries (€70k–€100k+) at network operators, regulators, and consultancies.

SALARY SIGNAL KEYWORDS TO USE

Load factor	Demand response	Peak shaving	Duck curve	Network peak demand
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